

NHANES Mini Project — Cleaned & Organized Version

1. Introduction

This project analyzes a small subset of NHANES-like demographic and anthropometric data.

We perform:

- Data cleaning
- Feature engineering (BMI, Age Groups)
- Statistical summaries
- Visualizations
- Outlier detection (Z-score)

2. Loading the Data

```
import pandas as pd
df=pd.read_csv('nhanes_clean.csv')
```

3. Creating Age Groups

```
bins = [0, 17, 40, 60, 80, 120]
labels = [
    '0-17 (child)',
    '18-40 (young)',
    '41-60 (adult)',
    '61-80 (senior)',
    '81+ (elder)'
]
df['AgeGroup']= pd.cut(df['Age'], bins=bins, labels=labels)
df.to_csv('nhanes_clean.csv', index= False)
```

4. Calculating BMI

```
df['BMI']=df['Weight']/(df['Height']/100)**2
```

5. Summary Statistics

Mean height & weight by age group

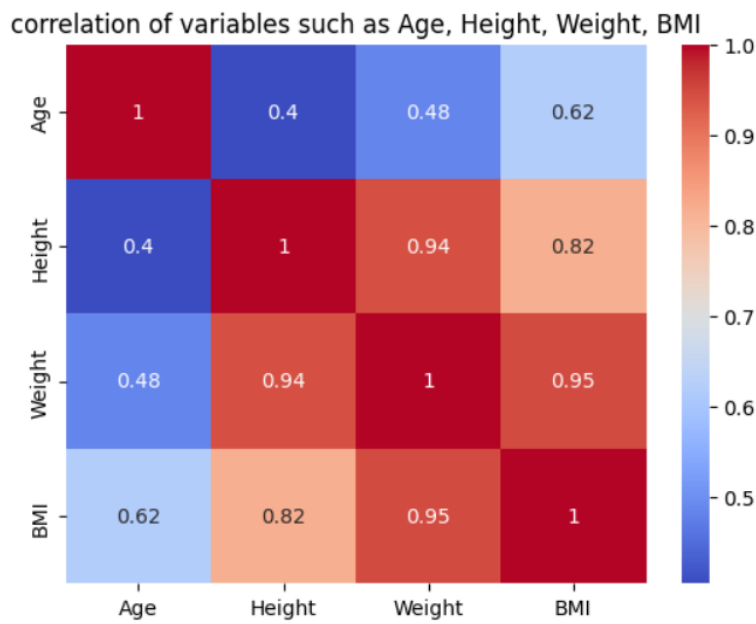
```
df.groupby("AgeGroup")[["Height", "Weight"]].mean()
```

Min & max values

```
df[["Height", "Weight"]].agg(["min", "max"])
```

6. Correlation Matrix

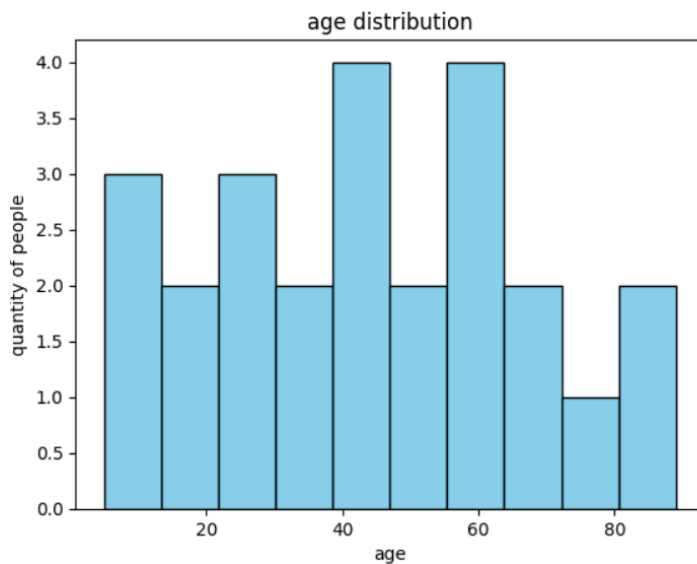
```
import seaborn as sns
import matplotlib.pyplot as plt
corr = df[["Age", "Height", "Weight", "BMI"]].corr()
sns.heatmap(corr, annot=True, cmap="coolwarm") plt.show()
```



7. Visualizations

Age distribution

```
plt.hist(df['Age'], bins=10, color='skyblue', edgecolor='black')
plt.title("age distribution")
plt.xlabel("age")
plt.ylabel("quantity of people")
plt.show()
```



Quantity of people by Age Group

```
sns.countplot(
    x='AgeGroup',
    data=df,
    color='red'
```

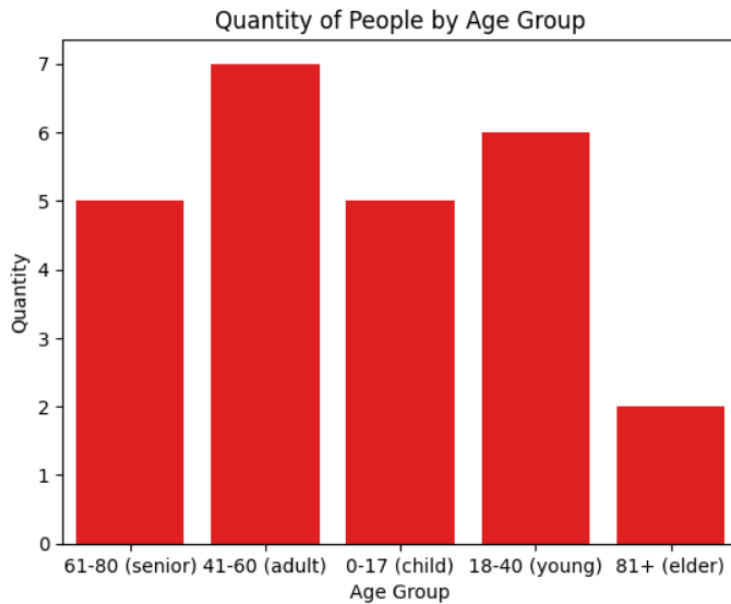
```
)
```

```
plt.title("Quantity of People by Age Group")
```

```
plt.xlabel("Age Group")
```

```
plt.ylabel("Quantity")
```

```
plt.show()
```



Quantity of Men and Women by Age Group

```
sns.countplot(
```

```
    x='AgeGroup',
```

```
    hue='Gender',
```

```
    data=df,
```

```
    palette='Set2'
```

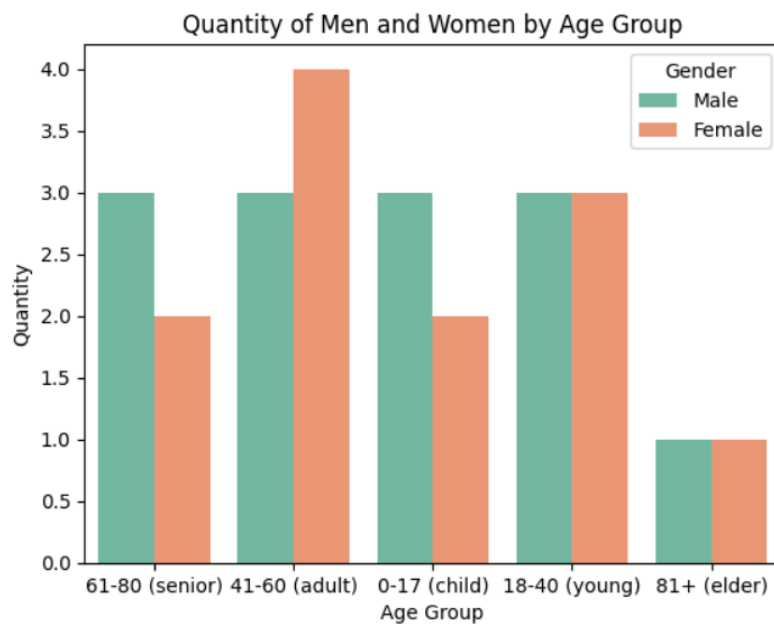
```
)
```

```
plt.title("Quantity of Men and Women by Age Group")
```

```
plt.xlabel("Age Group")
```

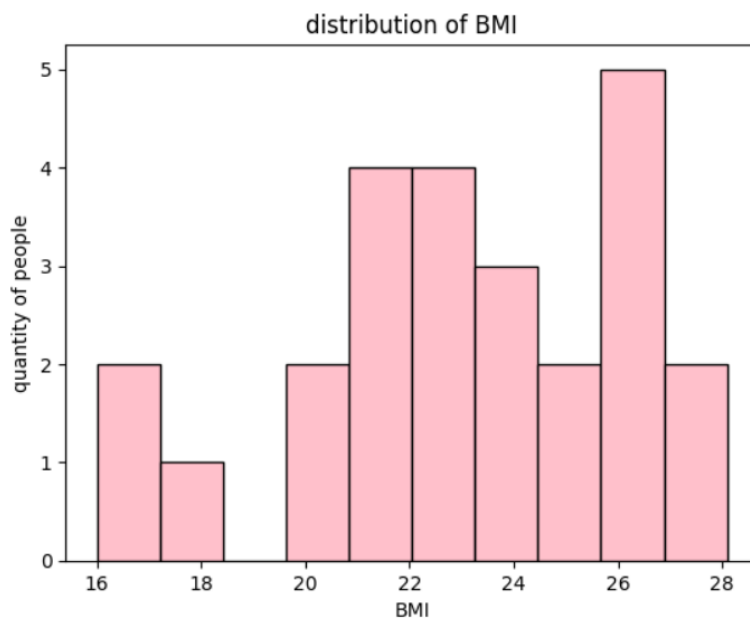
```
plt.ylabel("Quantity")
```

```
plt.show()
```



distribution of BMI

```
plt.hist(df['BMI'], bins=10, color='pink', edgecolor='black')
plt.title("distribution of BMI")
plt.xlabel("BMI")
plt.ylabel("quantity of people")
plt.show()
```



BMI by Age Group

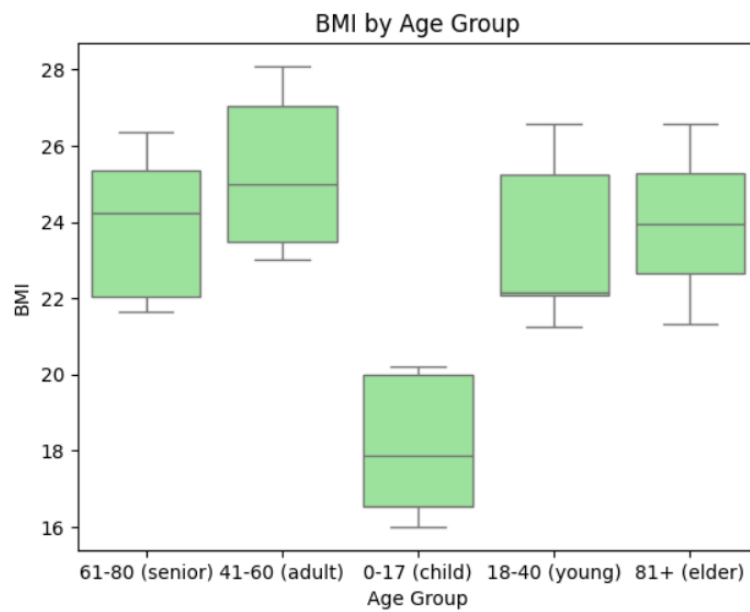
```
sns.boxplot(
    x='AgeGroup',
    y='BMI',
    data=df,
```

```

        color='lightgreen'
    )

plt.title("BMI by Age Group")
plt.xlabel("Age Group")
plt.ylabel("BMI")
plt.show()

```



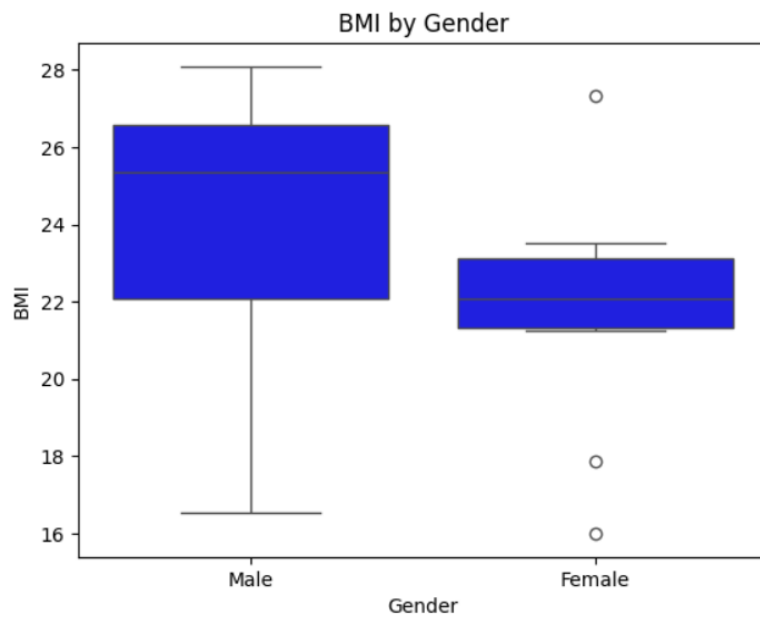
BMI by Gender

```

sns.boxplot(
    x='Gender',
    y='BMI',
    data=df,
    color='blue'
)

plt.title("BMI by Gender")
plt.xlabel("Gender")
plt.ylabel("BMI")
plt.show()

```



BMI by age group without outliers (z-score)

```
df['z_BMI']=(df['BMI']-df['BMI'].mean())/df['BMI'].std()
```

```
df_no_outliers=df[df['z_BMI'].abs()<3]
```

```
sns.boxplot(
```

```
    x='AgeGroup',
```

```
    y='BMI',
```

```
    hue='AgeGroup',
```

```
    data=df_no_outliers,
```

```
    palette='pastel',
```

```
    dodge=False
```

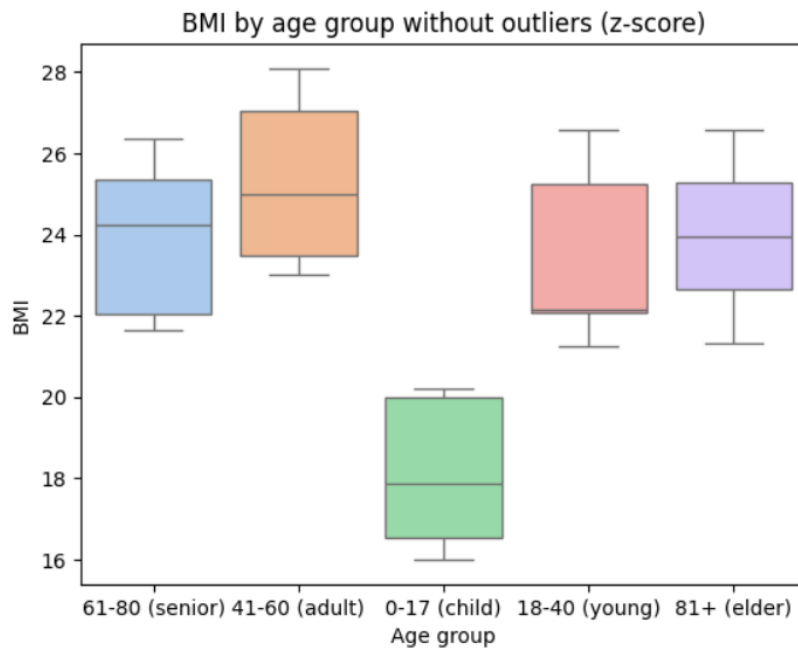
```
)
```

```
plt.title("BMI by age group without outliers (z-score)")
```

```
plt.xlabel("Age group")
```

```
plt.ylabel("BMI")
```

```
plt.show()
```



8. Outlier Removal (Z-Score)

```
df["z_BMI"] = (df["BMI"] - df["BMI"].mean()) / df["BMI"].std()
```

```
df_no_outliers = df[df["z_BMI"].abs() < 3]
```

Plot without outliers

```
sns.boxplot(x="AgeGroup", y="BMI", hue="AgeGroup",
            data=df_no_outliers, palette="pastel", dodge=False)
```

```
plt.title("BMI by Age Group (No Outliers)")
```

```
plt.show()
```

9. Conclusion

This project demonstrates:

- Cleaning and structuring health data
- Creating derived variables
- Statistical group comparisons